Course Section: Our Dynamic Universe – Gravitation

2022 Ph H2 Q5

(a)
Calculate the gravitational field strength at the surface of the Moon

Use the formula for gravitational field strength:
$g = \frac{GM}{r^2}$

Where:

- $G = 6.67 \times 10^{-11}\,\mathrm{m^3kg^{-1}s^{-2}}$
- $M = 7.35 \times 10^{22}\,\mathrm{kg}$ (mass of Moon)
- $r = 1.74 \times 10^{6}\,\mathrm{m}$ (radius of Moon)

$g = \frac{6.67 \times 10^{-11} \times 7.35 \times 10^{22}}{(1.74 \times 10^{6})^2}$

First, square the radius:
$(1.74 \times 10^{6})^2 = 3.03 \times 10^{12}$

Now compute:
$g = \frac{4.901 \times 10^{12}}{3.03 \times 10^{12}} = 1.62\,\mathrm{N\ kg^{-1}}$

(b)
Calculate the gravitational potential at the surface of the Moon

Use:
$V = -\frac{GM}{r}$

$V = -\frac{6.67 \times 10^{-11} \times 7.35 \times 10^{22}}{1.74 \times 10^{6}}$

$V = -\frac{4.901 \times 10^{12}}{1.74 \times 10^{6}} = -2.82 \times 10^{6}\,\mathrm{J\ kg^{-1}}$

(c)
Explain why the gravitational potential is negative

✅ Gravitational potential is defined as the work done **per unit mass** to move an object **from infinity to a point in the field.**

✅ Since gravity is **attractive**, it requires **negative work** (energy must be taken away) to move a mass from infinity to a point closer to the mass.

✅ Therefore, gravitational potential is always **negative** near a mass.

🧠 Revision Tips

- $g = \frac{GM}{r^2}$ gives gravitational field strength at a distance r from a mass M
- $V = -\frac{GM}{r}$ is always negative: more negative = deeper in the gravitational well
- Be careful with units and powers of ten when squaring or dividing large numbers
- Always explain negatives in gravitational potential in terms of energy required to move from infinity

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